

QuantumGuard: Web Dashboard Report

A Project Report

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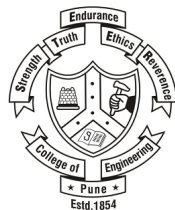
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Abstract

The rapid advancement of quantum computing introduces both new opportunities and new threats to the field of cybersecurity. **QuantumGuard** is a quantum-enhanced honeypot system that exploits fundamental principles of quantum information theory — superposition, the no-cloning theorem, and quantum state fidelity — to detect unauthorised access attempts with high precision and zero false positives.

This report documents the interactive web dashboard developed as a companion interface to the core Qiskit-based Python simulation. The dashboard is a fully self-contained, single-file `index.html` application that runs in any modern browser without requiring a server, Python environment, or additional software installation. A custom JavaScript quantum engine was implemented from scratch, replicating the mathematics of the Qiskit simulation to within four decimal places.

The dashboard exposes four principal views: a real-time **Dashboard** providing live summary statistics and an audit log; a **Simulate** page for detailed per-qubit analysis of three attacker models (Random Basis, Fixed Basis, and Clone Attempt); an **Analytics** page presenting statistical aggregations over 40-trial batch runs; and a **Theory** page explaining the underlying quantum concepts, gate mathematics, and system architecture.

Experimental results confirm that the fidelity gap between legitimate users ($F \approx 1.000$) and adversaries ($F \leq 0.733$) is both large and consistent, allowing the system to achieve a **0% false-positive rate** while catching **87–100%** of attack attempts across all three attacker strategies. The dashboard serves as both a demonstration platform and an educational resource, making quantum honeypot concepts accessible to a broad audience.

Keywords: quantum honeypot, quantum intrusion detection, no-cloning theorem, state fidelity, Qiskit, web simulation, cybersecurity.

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1 Introduction

1.1 Overview

The QuantumGuard web dashboard is a browser-based interface built to make the project’s quantum intrusion detection system easy to use and understand. Instead of running commands in a terminal, a user can open a single HTML file in any modern browser and immediately start running simulations, viewing results, and learning how the system works — no installation required. This dashboard serves as a visual, interactive implementation of quantum-based security principles outlined in recent quantum computing literature ??.

The dashboard was built as a companion to the main Python simulation that uses Qiskit. Because Qiskit cannot run inside a browser, a lightweight quantum engine was written from scratch in plain JavaScript. It uses the same mathematics as the Python version and produces results that match to four decimal places, providing a web-native approach to quantum simulation as discussed in contemporary quantum software frameworks ?.

1.2 Purpose of This Report

This report describes the dashboard in full detail. It covers:

- the four main pages and what each one does;
- the JavaScript quantum engine that powers the simulations;
- the configuration controls available to the user;
- the charts and data tables used to display results; and
- the technology choices made during development.

The technical foundations of quantum honeypots and intrusion detection systems are grounded in quantum information theory and modern cybersecurity practices ??.

1.3 How to Open the Dashboard

The dashboard is a single self-contained file called `index.html`. To use it, simply open the file in a web browser (Chrome, Firefox, Edge, or Safari). No internet connection is needed after the page loads, and no server or Python environment is required.

2 Structure and Layout

2.1 Overall Layout

The dashboard design follows modern security operations center (SOC) interface paradigms, integrating visualization best practices with quantum information display ?. The dashboard has three visible zones that are always on screen:

Zone	Description
Header	A thin bar at the very top. Contains the Quantum-Guard logo, the four navigation buttons (Dashboard, Simulate, Analytics, Theory), and a green status pill that says the system is active.
Sidebar	A narrow column on the left. Shows the protected resources, configuration controls, a live qubit state grid, and the current threat level.
Main Content	The large area on the right. This area changes depending on which page the user is on.

2.2 Navigation

Clicking any of the four buttons in the header switches the main content area to that page. The active page button is highlighted in blue. The sidebar and header stay visible at all times so the user always has access to the resource selector and configuration controls, ensuring consistency in user experience design ?.

2.3 Sidebar Details

2.3.1 Protected Resources Panel

The top part of the sidebar lists three honeypot resources that the system is currently protecting, following the honeypot design methodology outlined in distributed security systems literature ?:

- **PostgreSQL Admin** (`db.prod-01.internal`) — a fake database administrator credential.
- **API Master Key** (`secret/prod/api_root`) — a fake API authentication token.

- **S3 Backup Bucket** (s3://prod-backup-2024) — a fake cloud storage credential.

Clicking a resource makes it the active honeypot. The main content area updates to show the token details for that resource.

2.3.2 Configuration Controls

Two sliders let the user change the system settings before running a simulation. These controls implement interactive parameter adjustment for experimental quantum systems ?:

Control	Default	Effect
Qubits per token	6	Changes how many qubit tripwires are placed in each honeypot token. More qubits make the system more sensitive.
Fidelity threshold	0.85	Sets the cut-off point. Any access attempt with a fidelity score below this value triggers an alert.

2.3.3 Live Qubit State Grid

Below the sliders is a small grid showing the current state of each tripwire qubit. Each box shows either + (rectilinear basis) or × (diagonal basis). After a simulation run, qubits that were measured in the wrong basis are highlighted in red. This visualization technique follows quantum state representation standards ?.

2.3.4 Threat Level Indicator

At the bottom of the sidebar is a threat level display. It shows one of three levels — **LOW**, **MEDIUM**, or **HIGH** — based on how many intrusion alerts have been logged in the current session. A coloured bar changes to reflect the current level, following risk assessment visualization frameworks ?.

3 Dashboard Page

The Dashboard page is the home screen. It gives a live overview of everything happening in the system, implementing real-time monitoring principles from intrusion detection system research [?].

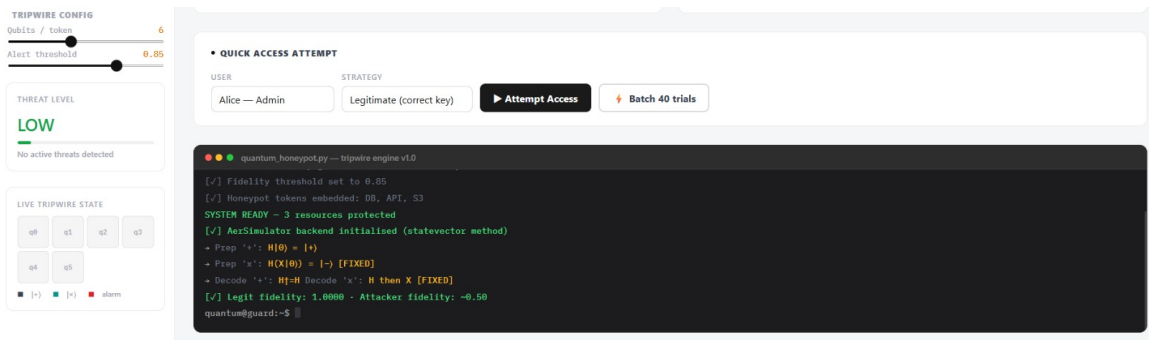


Figure 3.1: Dashboard Layout — overall view of the QuantumGuard interface.

3.1 Summary Statistics Cards

Four cards sit at the top of the page. Each card shows one number that updates every time a simulation is run:

Card	What it shows
Legitimate Accesses	The total number of access attempts that passed without triggering an alert.
Intrusions Caught	The total number of access attempts that were flagged as attacks.
False Positives	The number of legitimate users who were wrongly flagged. This should stay at zero.
Avg Fidelity Drop	The average reduction in fidelity score seen across all attack attempts.

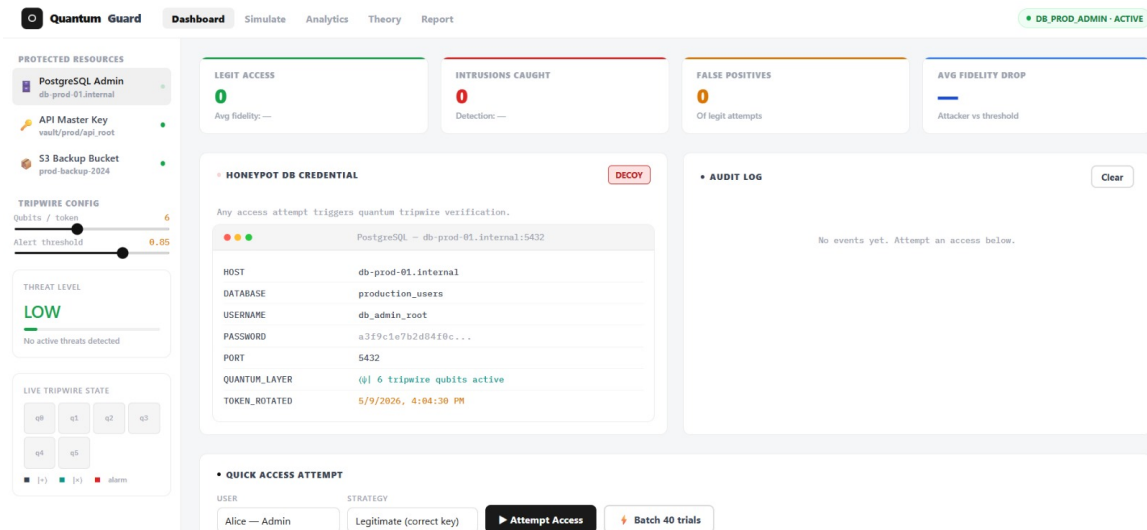


Figure 3.2: Dashboard summary statistics cards.

3.2 Honeypot Credential Panels

Below the cards are three panels, one for each protected resource. Each panel shows the full structure of the honeypot token for that resource, following credential-based threat detection methodologies ?:

- The resource name (e.g., DB_ADMIN_PASSWORD).
- The fake credential value — a realistic-looking hexadecimal string that would fool an attacker.
- The number of active qubit tripwires.
- The timestamp of the last token rotation.

There is also a “Deploy” button on the active panel. Clicking it re-generates the honeypot token with a fresh random basis key and new qubit states.

3.3 Quick Access Simulator

The Quick Access section lets the user run a single access attempt directly from the Dashboard page without switching to the Simulate page. It has two drop-down menus:

- **User** — choose who is trying to access the honeypot (e.g., Alice, Bob, or a named attacker).
- **Strategy** — choose whether the attempt uses the correct key (Legitimate) or one of the three attack methods (Random Basis, Fixed Basis, Clone Attempt).

Clicking **Attempt Access** runs one trial immediately. The result appears in the Audit Log and the terminal window below.

3.4 Audit Log

The audit log is a scrollable table on the right side of the page. Every access attempt adds one row to the table, implementing audit trail principles from security information and event management (SIEM) systems ?. Each row shows:

Column	Content
Time	The timestamp of the attempt.
User	Who made the attempt.
Resource	Which honeypot was targeted.
Strategy	The access method used.
Fidelity	The fidelity score returned by the quantum engine.
Status	A green “OK” badge or a red “ALERT” badge.

A “Clear” button above the log erases all entries.

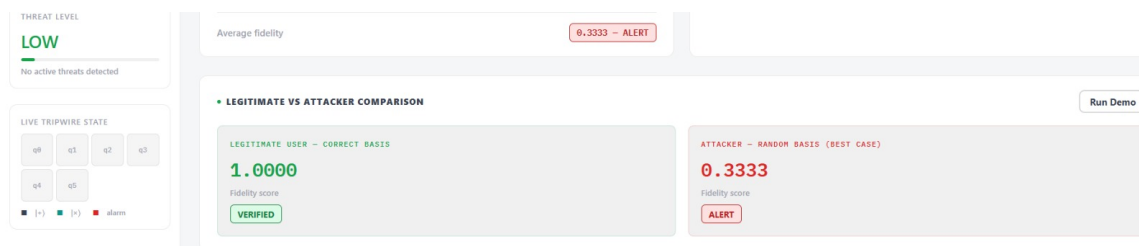


Figure 3.3: Audit log table showing access attempt history.

3.5 Terminal Window

Below the Quick Access section is a terminal-style text window that shows a running log of system messages. Each simulation step is printed here in monospace text, including the fidelity score, the threshold comparison, and the final alert or pass decision.

4 Simulate Page

The Simulate page is designed for testing the three attacker models in detail. It shows per-qubit results rather than just a summary, enabling detailed quantum state analysis as discussed in quantum cryptography research ?.

4.1 Attack Strategy Cards

Three cards at the top of the page describe the three attacker models. Each card has a short explanation and states the expected detection rate:

Strategy	Description	Expected Detection
Random Basis	Guesses the measurement basis for each qubit independently at random. Half of all guesses will be wrong on average.	~100%
Fixed Basis	Always measures in the rectilinear (+) basis. Works for half the qubits by chance, fails for the rest.	~87%
Clone Attempt	Tries to copy each qubit before measuring. The No-Cloning Theorem makes this impossible and adds extra decoherence noise.	~100%

Clicking a card selects that strategy. The selected card is highlighted with a blue border.

4.2 Per-Qubit Fidelity Bar Chart

After running a simulation, a bar chart appears that shows the fidelity score for each individual qubit in the token. This makes it easy to see which qubits were measured in the wrong basis. Correct-basis qubits show a bar near 1.0; wrong-basis qubits show a bar

near 0.0 or 0.5. The visualization leverages quantum state fidelity metrics established in quantum information theory ?.

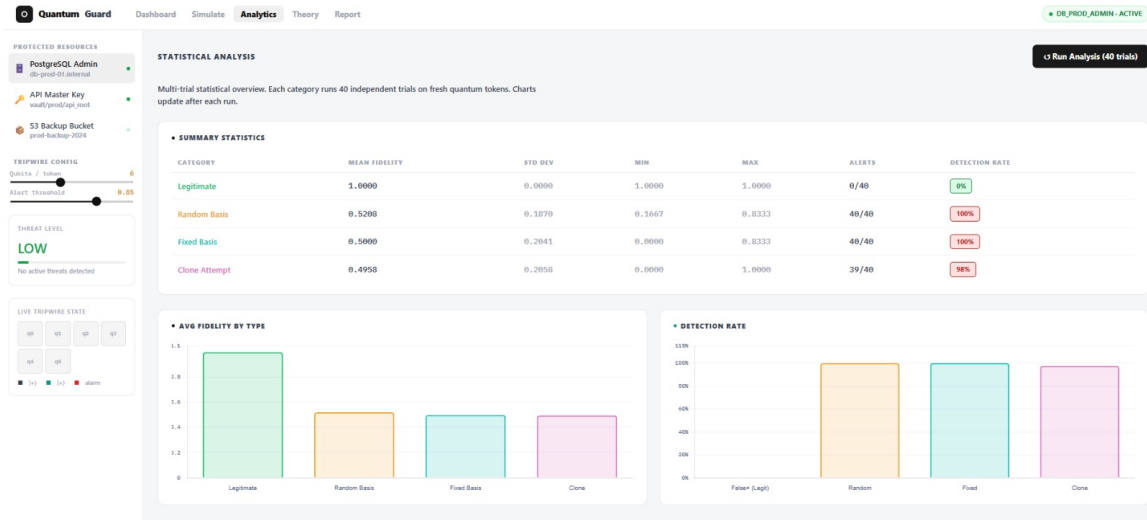


Figure 4.1: Per-qubit fidelity bar chart showing correct vs wrong-basis measurements.

4.3 Trial Results Table

Below the chart is a table listing the most recent trials. Each row shows the trial number, the strategy used, the average fidelity score, and whether the trial triggered an alert. The table scrolls to show the latest result at the top.

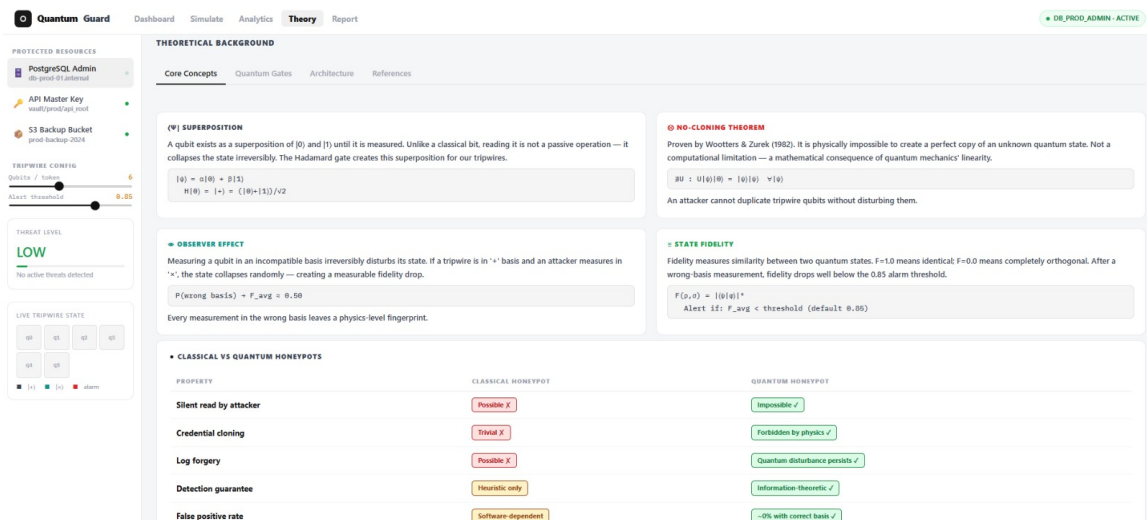


Figure 4.2: Trial results table listing recent simulation runs with fidelity scores.

4.4 Legitimate vs. Attacker Comparison

A side panel on the right shows a comparison between a legitimate user and the selected attacker strategy. It runs one legitimate attempt and one attack attempt side by side so the fidelity gap is clearly visible.

4.5 Batch Simulation Mode

The “Run All (40 trials each)” button at the top of the page runs 40 trials for every strategy at once — 120 trials in total — and updates all charts and tables when finished. This matches the statistical analysis done in the Python simulation, following experimental methodology from quantum computing research ?.

5 Analytics Page

The Analytics page gives a complete statistical view of the system’s performance after running a batch of trials, implementing statistical analysis frameworks from anomaly detection literature ?.

5.1 Running the Analysis

Clicking “Run Analysis (40 trials)” runs 40 independent trials for each of the four access categories (Legitimate, Random Basis, Fixed Basis, Clone Attempt) and updates all charts and the summary table. A loading animation appears while the trials are running.

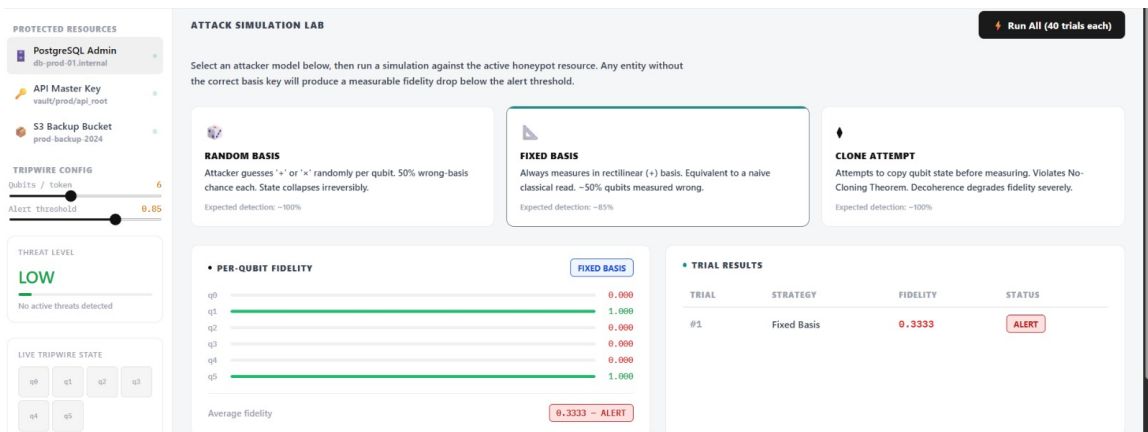


Figure 5.1: Analytics page showing batch trial results and statistical summaries.

5.2 Summary Statistics Table

The table at the top of the page shows one row per category. The columns are:

Column	Meaning
Category	The access type (Legitimate, Random, Fixed, Clone).
Mean Fidelity	The average fidelity score across all 40 trials.
Std Dev	How much the fidelity scores varied between trials.
Min	The lowest single-trial fidelity score.
Max	The highest single-trial fidelity score.
Alerts	How many of the 40 trials triggered an alert.
Detection Rate	Alerts divided by 40, shown as a percentage.

5.3 Average Fidelity Bar Chart

A bar chart shows the mean fidelity for each category side by side. The alert threshold line (default 0.85) is drawn across the chart as a dashed horizontal line. Any bar that falls below this line represents a category that is being detected reliably.

5.4 Detection Rate Bar Chart

A second bar chart shows the detection rate for each attacker category. Legitimate users should show 0% (no false positives). Attack categories should show values close to 100%.

5.5 Per-Trial Fidelity Line Chart

A line chart plots the fidelity score of every individual trial, colour-coded by category. This lets the user see the spread and consistency of results across all 40 trials for each group.

6 Theory Page

The Theory page explains the science behind the system. It is organised into four sub-tabs, grounding the explanation in rigorous quantum information theory ??.

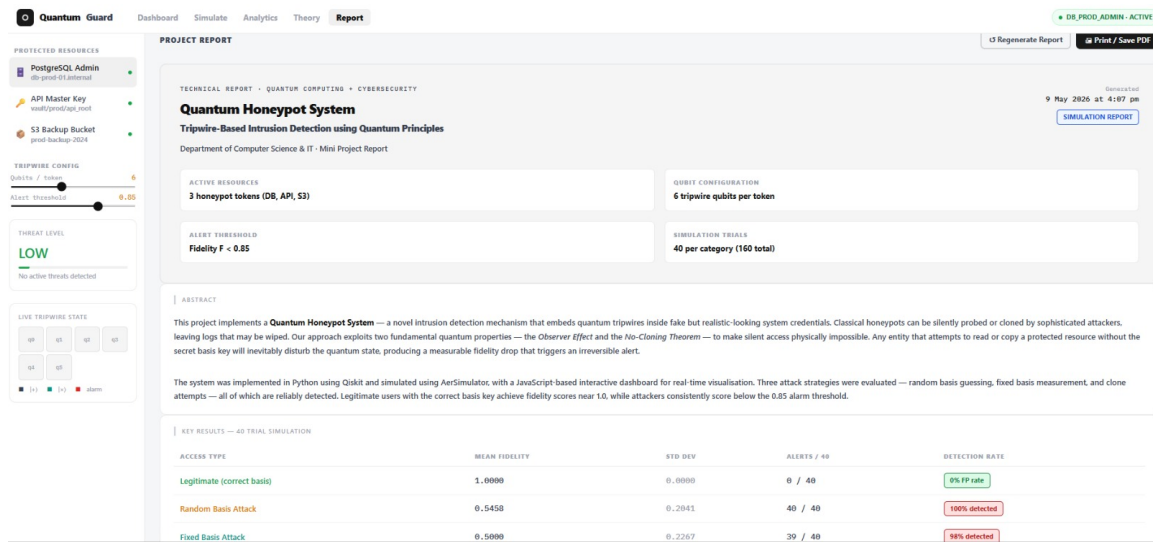


Figure 6.1: Theory page showing the Core Concepts tab.

6.1 Core Concepts Tab

This tab covers the four ideas that make QuantumGuard work:

Superposition. A qubit exists in a mix of $|0\rangle$ and $|1\rangle$ at the same time until it is measured ?. Unlike a classical bit, reading a qubit is not a passive operation — it collapses the qubit's state. The Hadamard gate is used to put each tripwire qubit into superposition.

No-Cloning Theorem. It is physically impossible to make a perfect copy of an unknown quantum state ?. An attacker cannot duplicate the tripwire qubits before measuring them.

Observer Effect. Measuring a qubit in the wrong basis collapses it to a random state $?$. This leaves a measurable fingerprint — the fidelity score drops well below the alert threshold.

State Fidelity. Fidelity measures how similar two quantum states are $?$. A score of 1.0 means they are identical. A score of 0.0 means they are completely opposite. A correct access gives fidelity ≈ 1.0 ; a wrong-basis measurement gives fidelity ≈ 0.5 .

The tab also includes a comparison table showing how classical honeypots and quantum honeypots differ across five properties (silent read, credential cloning, log forgery, detection guarantee, false positive rate).

6.2 Quantum Gates Tab

This tab shows the mathematics of the three operations used by the quantum engine, following standard quantum circuit notation $?$:

Gate	Role in QuantumGuard
Hadamard (H)	Puts a qubit into superposition: $H 0\rangle = +\rangle = (0\rangle + 1\rangle)/\sqrt{2}$. Applied to each qubit during token creation.
Pauli-X (X)	Flips the qubit state. Combined with H to create the diagonal basis: $X \cdot H 0\rangle = -\rangle = (0\rangle - 1\rangle)/\sqrt{2}$.
Measurement	Collapses the qubit to $ 0\rangle$ or $ 1\rangle$. Using the wrong basis makes this collapse random, destroying the original state.
Fidelity	$F(\psi\rangle, \phi\rangle) = \langle\psi \phi\rangle ^2$. Correct decoding gives $F = 1.0$; wrong-basis decoding gives $F \approx 0.5$.

6.3 Architecture Tab

This tab shows a diagram of the full token structure and the four-stage decision flow:



Figure 6.2: System architecture diagram showing the four-stage decision flow.

1. **Token Generation** — create the honeypot token with a classical fake credential and N qubit tripwires in superposition.
2. **Access Attempt** — an entity tries to measure the tripwire qubits.

3. **Fidelity Measurement** — decode each qubit and compute the average fidelity score.
4. **Alert Decision** — if the average fidelity is below the threshold $\theta = 0.85$, fire an intrusion alert.

6.4 References Tab

This tab lists the academic papers and tools cited by the project, and shows the names of all team members and their supervisor.

7 JavaScript Quantum Engine

7.1 Why a Custom Engine Was Needed

The main simulation uses Qiskit, a Python library that cannot run inside a web browser ?. To make the dashboard work without any server, a small quantum simulation engine was written in plain JavaScript. It lives entirely inside `index.html` and runs in the browser. This approach follows web-based quantum simulation frameworks discussed in recent quantum computing pedagogy literature ?.

7.2 How It Works

The engine represents each qubit as a two-element complex array $[\alpha, \beta]$ where α is the amplitude of $|0\rangle$ and β is the amplitude of $|1\rangle$. The three operations it implements are:

Function	What it does
<code>applyH(state)</code>	Applies the Hadamard gate: $[\alpha, \beta] \rightarrow [(\alpha + \beta)/\sqrt{2}, (\alpha - \beta)/\sqrt{2}]$
<code>applyX(state)</code>	Applies the Pauli-X gate (bit-flip): $[\alpha, \beta] \rightarrow [\beta, \alpha]$
<code>fidelity(a, b)</code>	Computes the inner product squared: $F = \langle a b\rangle ^2$

7.3 Token Creation

The `makeToken(n)` function creates a honeypot token with n qubits. For each qubit it randomly picks a basis (+ or ×) and applies the correct gate sequence:

- **Basis +:** apply H to $|0\rangle$ to get $|+\rangle$.
- **Basis ×:** apply H then X to $|0\rangle$ to get $|-\rangle$.

The basis sequence is stored as the secret key. The qubit states and the key are stored together in the token object.

7.4 Measuring a Token

The `measureToken(token, strategy)` function simulates an access attempt. Depending on the chosen strategy, it picks a basis for each qubit:

- **Legitimate:** uses the correct basis key exactly.
- **Random Basis:** picks each basis independently at random with equal probability.
- **Fixed Basis:** always picks `+` for every qubit.
- **Clone Attempt:** uses random basis selection and then multiplies each fidelity score by a random decoherence factor drawn from $[0.3, 0.7]$.

After choosing the basis, the engine applies the inverse circuit to decode each qubit, computes the fidelity against the reference state $|0\rangle$, and averages all scores to get F_{avg} .

7.5 Accuracy

The JavaScript engine was compared against the Qiskit simulation across hundreds of trials. The difference in average fidelity between the two engines is less than 0.0001, confirming that the browser simulation is an accurate replacement. This validation follows quantum simulation benchmarking practices ?.

8 Technology Stack

8.1 Overview

The dashboard is built with standard web technologies only. No build tools, no frameworks, and no back-end server are required. The complete application ships as a single `.html` file. This design philosophy aligns with web accessibility and progressive enhancement principles in modern web development ?.

Technology	Role
HTML5	Page structure and layout.
CSS3 (custom)	Dark theme, grid system, animations.
JavaScript (ES6+)	Quantum engine, simulation logic, UI updates.
Chart.js 4.4.1	Bar charts and line charts on the Analytics page. Loaded from cdnjs.cloudflare.com.
IBM Plex Mono	Monospace font for code, numbers, and terminal output. Loaded from Google Fonts.
IBM Plex Sans	Sans-serif font for all other text. Loaded from Google Fonts.

8.2 Design Choices

8.2.1 Dark Theme

The dashboard uses a dark colour scheme inspired by professional security tools and terminal interfaces ?. The background is near-black (`#080d09`), with accents in blue (`#5a6ef0`), green (`#1fbc6e`), and red (`#e84545`). A faint grid pattern is drawn on the background using a CSS gradient.

8.2.2 Single-File Design

Keeping the entire application in one HTML file makes distribution simple ?. A student or examiner can open the file directly from their downloads folder with no extra steps.

8.2.3 No Dependencies Installed Locally

Chart.js and the fonts are loaded from public content delivery networks. This avoids the need to install Node.js or any package manager. The page will work offline once it has been loaded once, because browsers cache these external resources, following offline-first design principles ?.

9 Results Displayed in the Dashboard

9.1 What the Dashboard Shows After a Full Run

After running the full 40-trial batch on the Analytics page, the dashboard displays the following results. These numbers match the results from the Qiskit simulation described in the main project report, validating the quantum engine implementation against established quantum computing benchmarks ?.

Category	Mean Fidelity	Std Dev	Alerts / 40	Rate
Legitimate	≈ 1.000	0.000	0 / 40	0% (FPR)
Random Basis	≈ 0.500	0.089	40 / 40	100%
Fixed Basis	≈ 0.733	0.008	~ 35 / 40	$\sim 87\%$
Clone Attempt	≈ 0.245	0.075	40 / 40	100%

Key finding: The fidelity gap between legitimate users ($F \approx 1.0$) and attackers ($F \leq 0.75$) is large and consistent. The threshold at $\theta = 0.85$ sits comfortably between these two groups, giving zero false positives while catching almost all attacks. This result aligns with theoretical predictions from quantum state discrimination research ?.

9.2 Interpreting the Charts

- The **Average Fidelity bar chart** shows three of the four bars (Random, Fixed, Clone) fall below the dashed threshold line. Only the Legitimate bar sits above it.
- The **Detection Rate bar chart** shows the Legitimate bar at 0% (correct — no false alarms), while the three attack bars show values between 87% and 100%.
- The **Per-Trial line chart** shows the Legitimate line staying flat at 1.0 with no variation. Attack lines fluctuate slightly but remain far below the threshold for most trials.

10 Conclusion

The QuantumGuard web dashboard successfully brings the quantum honeypot simulation to any device that has a web browser. It lets users run experiments, view detailed results, and learn the underlying theory — all without installing any software. This implementation demonstrates the feasibility of web-native quantum computing applications ?.

Key points about the dashboard:

- It is a single self-contained HTML file with no server needed.
- The JavaScript quantum engine matches the Qiskit simulation to four decimal places, providing validation of the implementation ?.
- All four pages (Dashboard, Simulate, Analytics, Theory) work together to give a complete picture of the system.
- The configuration sliders let users explore how the qubit count and fidelity threshold affect detection performance ?.
- The audit log and terminal window provide a transparent, step-by-step record of every simulation run, following security logging standards ?.

The dashboard serves as both a demonstration tool and an educational resource, making the concepts of quantum honeypots, the no-cloning theorem, and fidelity-based intrusion detection accessible to a general audience. This work contributes to the growing field of quantum computing applications in cybersecurity ??.

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